WORKED SOLUTION

STEP III 2022, Question 5

Pure • Symmetric Definite Integrals

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Read the question before you go on. It is in the STEP III paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

The key idea

On a symmetric interval $[-a, a]$, the substitution $x \mapsto -x$ lets us fold the integral onto $[0, a]$, pairing the value at x with the value at $-x$. For integrands of the form $1/(1 + \text{something})$, the pair $\frac{1}{1+p} + \frac{1}{1+q}$ collapses to exactly 1 precisely when $pq = 1$, and $\mathrm{e}^x \cdot \mathrm{e}^{-x} = 1$ is the model case. The whole question is about spotting that pairing and then knowing when it is the *only* way the answer can come out to be a.

Solution

Part (i)

Let $I = \displaystyle\int_{-a}^{a} \frac{1}{1+\mathrm{e}^x}\,\mathrm{d}x$. Substituting $x = -u$ (so $\mathrm{d}x = -\mathrm{d}u$, and the limits swap back),

$$I = \int_{-a}^{a} \frac{1}{1+\mathrm{e}^{-u}}\,\mathrm{d}u = \int_{-a}^{a} \frac{\mathrm{e}^u}{\mathrm{e}^u+1}\,\mathrm{d}u,$$

after multiplying top and bottom by e^u. Adding this to the original form of I,

$$2I = \int_{-a}^{a} \frac{1+\mathrm{e}^x}{1+\mathrm{e}^x}\,\mathrm{d}x = \int_{-a}^{a} 1\,\mathrm{d}x = 2a,$$

so $I = a$ for every $a \geq 0$, as required.

Part (ii)

The explanation. Define $G(a) = \displaystyle\int_0^a g(x)\,\mathrm{d}x$ for $a \geq 0$. Since g is continuous, the fundamental theorem of calculus says G is differentiable with $G'(a) = g(a)$ (a one-sided derivative at $a = 0$). But G is identically zero on $[0, \infty)$, so its derivative is zero there. Hence $g(a) = 0$ for all $a \geq 0$. (Continuity matters: without it, g could be non-zero at isolated points without affecting any integral.)

Folding the integral. Split at 0 and put $x = -u$ in the left half:

$$\int_{-a}^{0} \frac{\mathrm{d}x}{1+\mathrm{f}(x)} = \int_{0}^{a} \frac{\mathrm{d}u}{1+\mathrm{f}(-u)}.$$

Writing $a = \int_0^a 1\,\mathrm{d}x$, we get

$$\int_{-a}^{a} \frac{\mathrm{d}x}{1+\mathrm{f}(x)} - a = \int_{0}^{a} g(x)\,\mathrm{d}x, \qquad \text{where} \quad g(x) = \frac{1}{1+\mathrm{f}(x)} + \frac{1}{1+\mathrm{f}(-x)} - 1.$$

Because $\mathrm{f} \geq 0$, both denominators are at least 1, so g is continuous.

> **?** **Ask yourself** The result just explained is a one-way statement. Which direction of the "if and only if" does it give, and where does the other come from?

- If $g(x) = 0$ for all $x \geq 0$, then $\int_0^a g = 0$ and the integral equals a for every $a \geq 0$. This direction is immediate.
- Conversely, if the integral equals a for every $a \geq 0$, then $\int_0^a g = 0$ for every $a \geq 0$, and the first paragraph (with this continuous g) gives $g(x) = 0$ for all $x \geq 0$.

So the integral condition holds if and only if $g(x) = 0$ for all $x \geq 0$.

The product condition. Write $p = \mathrm{f}(x)$, $q = \mathrm{f}(-x)$. Multiplying through by $(1+p)(1+q) > 0$,

$$\begin{aligned} \frac{1}{1+p} + \frac{1}{1+q} = 1 &\iff (1+q) + (1+p) = (1+p)(1+q) \\ &\iff 2 + p + q = 1 + p + q + pq \\ &\iff pq = 1. \end{aligned}$$

So $g(x) = 0$ for all $x \geq 0$ if and only if $\mathrm{f}(x)\mathrm{f}(-x) = 1$ for all $x \geq 0$. Finally, the expression $\mathrm{f}(x)\mathrm{f}(-x)$ is unchanged when x is replaced by $-x$, so it equals 1 for all $x \geq 0$ exactly when it equals 1 for all real x. That completes the chain of equivalences.

Part (iii)

Fold the integral exactly as before, now using $\mathrm{h}(-u) = \mathrm{h}(u)$ in the left half:

$$\begin{aligned} \int_{-a}^{a} \frac{\mathrm{h}(x)}{1+\mathrm{f}(x)}\,\mathrm{d}x &= \int_{0}^{a} \frac{\mathrm{h}(x)}{1+\mathrm{f}(x)}\,\mathrm{d}x + \int_{0}^{a} \frac{\mathrm{h}(-u)}{1+\mathrm{f}(-u)}\,\mathrm{d}u \\ &= \int_{0}^{a} \mathrm{h}(x) \left(\frac{1}{1+\mathrm{f}(x)} + \frac{1}{1+\mathrm{f}(-x)} \right) \mathrm{d}x \\ &= \int_{0}^{a} \mathrm{h}(x)\,\mathrm{d}x, \end{aligned}$$

because the bracket equals 1 whenever $\mathrm{f}(x)\mathrm{f}(-x) = 1$, by the equivalence shown in part (ii). (The conditions $\mathrm{f} \geq 0$ and continuity just make every integrand continuous.)

Part (iv)

? **Ask yourself** How can $\dfrac{\mathrm{e}^{-x}}{\cosh x}$ be written in the form $\dfrac{\text{const}}{1+\mathrm{f}(x)}$?

Since $\cosh x = \frac{1}{2}(\mathrm{e}^{x}+\mathrm{e}^{-x})$,

$$\frac{\mathrm{e}^{-x}\cos x}{\cosh x} = \frac{2\mathrm{e}^{-x}\cos x}{\mathrm{e}^{x}+\mathrm{e}^{-x}} = \frac{2\cos x}{1+\mathrm{e}^{2x}},$$

multiplying top and bottom by e^{x}. Take $\mathrm{f}(x) = \mathrm{e}^{2x}$, which is continuous, positive, and satisfies $\mathrm{f}(x)\mathrm{f}(-x) = \mathrm{e}^{2x}\mathrm{e}^{-2x} = 1$; and take $\mathrm{h}(x) = 2\cos x$, which is continuous and even. Part (iii) with $a = \frac{1}{2}\pi$ gives

$$\int_{-\frac{1}{2}\pi}^{\frac{1}{2}\pi} \frac{\mathrm{e}^{-x}\cos x}{\cosh x}\,\mathrm{d}x = \int_{0}^{\frac{1}{2}\pi} 2\cos x\,\mathrm{d}x = \Big[2\sin x\Big]_{0}^{\frac{1}{2}\pi} = 2.$$

Checks

- Part (i) is the special case $\mathrm{f}(x) = \mathrm{e}^{x}$, $\mathrm{h} \equiv 1$ of part (iii), and gives $\int_0^a 1\,\mathrm{d}x = a$, as it should.
- A non-example: $\mathrm{f}(x) = x^2$ has $\mathrm{f}(x)\mathrm{f}(-x) = x^4 \neq 1$, and indeed $\int_{-1}^{1} \frac{\mathrm{d}x}{1+x^2} = \frac{\pi}{2} \neq 1$.
- Evaluating the integral in part (iv) numerically with Simpson's rule gives 2.0000000; the identity in part (iii) was also confirmed by computer for many random choices of $\mathrm{f} = \mathrm{e}^{(\text{odd function})}$ and even h.

Common slips

- Proving only one direction of the "if and only if" in part (ii): the step from the integral condition back to $g = 0$ is exactly where the first result (and the continuity of g) is needed.
- Forgetting to say why $\mathrm{f}(x)\mathrm{f}(-x) = 1$ for $x \geq 0$ gives it for all x: the expression is symmetric under $x \mapsto -x$.
- In part (iv), trying to integrate $\mathrm{e}^{-x}\cos x/\cosh x$ directly instead of rewriting it as $2\cos x/(1+\mathrm{e}^{2x})$.

Beyond STEP

The argument in part (ii) is the fundamental theorem of calculus used "backwards", and it is the reason continuity hypotheses appear so often in analysis: integrals cannot see what happens at individual points, so only continuity lets you recover a function's values from its integrals. The folding trick is a cousin of splitting a function into its even and odd parts; here $\frac{1}{1+\mathrm{f}(x)} - \frac{1}{2}$ is odd exactly when $\mathrm{f}(x)\mathrm{f}(-x) = 1$. Abbott's

Understanding Analysis gives a careful treatment of the fundamental theorem and why continuity is needed.

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